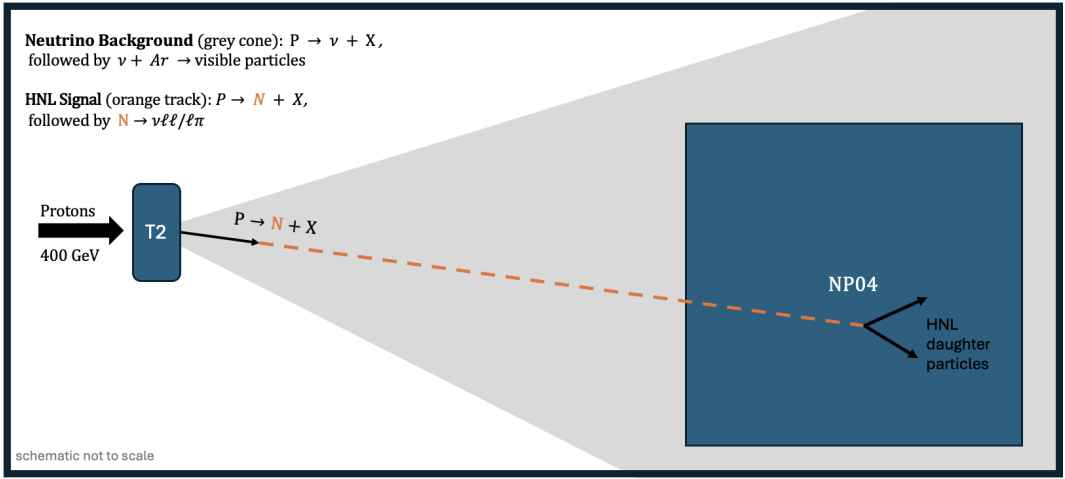
HNL Sensitivity Analysis for ProtoDUNE

NP04 Truth-Level

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NP04 provides a large LAr target for visible HNL decays



Aim and Analysis Channels

Aim: Identify GENIE neutrino backgrounds that mimic HNL signal topologies, suppress them using kinematic selections, and calculate the background-inclusive sensitivity to $|U_{\mu N}|^2$.

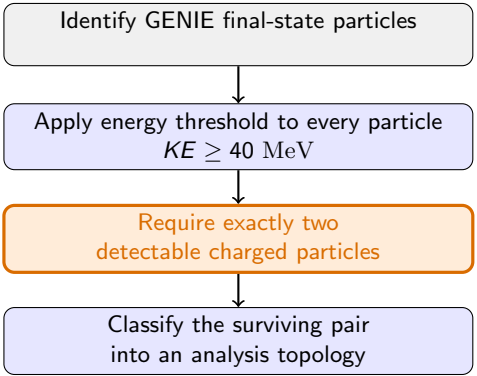
Topology	HNL decay channels	Parent mesons
2μ-**like**	$N \to \nu\mu^+\mu^-, \quad N \to \mu^\pm\pi^\mp$	$K^\pm,\ D^\pm,\ D_s^\pm$
1μ-**like** $+1e$	$N \to \nu e^\pm\mu^\mp$	$K^\pm,\ D^\pm,\ D_s^\pm$
$2e$	$N \to \nu e^+e^-$	$\pi^\pm,\ K^\pm,\ D^\pm,\ D_s^\pm$

Signal: Latest HNLux $U_{\mu N}$ and U_{eN} mixing samples.

Background: GENIE $\nu_e, \bar{\nu}_e, \nu_\mu, \bar{\nu}_\mu$ interactions, normalised to 1.75×10^{19} POT.

Truth level only: no G4 propagation or detector reconstruction.

Background Topology Selection Procedure



- Extra visible protons and hadronic activity are used to veto a large number of events.
- Neutrons and neutrinos above the visible energy threshold are not used to veto events, owing to lack of detector signature.
- 7.86% of background events remain after topological selection

Boosted HNL Kinematics Motivate Mass-Dependent Cuts

HNLs reaching NP04 are forward-going and boosted. Their visible decay products are typically collimated and have low combined transverse momentum.

$$E_{\mathrm{pair}} = E_1 + E_2$$

Visible-pair energy

$$p_{T,\mathrm{pair}} = |(\mathbf{p}_1 + \mathbf{p}_2)_T|$$

Pair transverse momentum

$$\theta_{\mathrm{pair}} = \angle(\mathbf{p}_1, \mathbf{p}_2)$$

Opening angle

$$E_{\mathrm{pair}} < E_{\mathrm{pair}}^{\mathrm{max}}, \qquad p_{T,\mathrm{pair}} < p_{T,\mathrm{pair}}^{\mathrm{max}}, \qquad \theta_{\mathrm{pair}} < \theta_{\mathrm{pair}}^{\mathrm{max}}.$$

Cut optimisation

$$\underset{\mathrm{cuts}}{\mathrm{maximize}} \left(\frac{\epsilon_S}{\epsilon_B} \right), \qquad \epsilon_S \geq 0.85.$$

Cuts are optimised jointly at each HNL mass and topology using weighted signal and background yields.

Example cut performance: 2-mu topology 1018 MeV

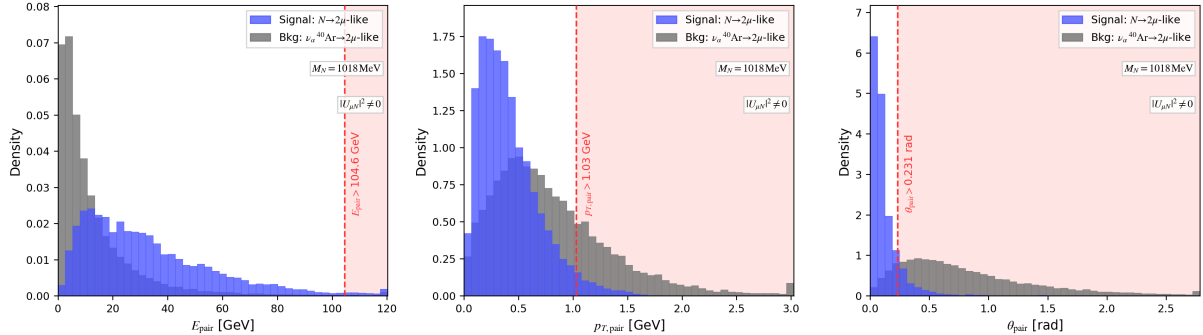


Figure: Optimized kinematic cuts obtained from grid scan over upper thresholds on $E_{\rm pair}$, $p_{T,\rm pair}$, and $\theta_{\rm pair}$. $\epsilon_S = 0.86$, $1 - \epsilon_B^{\rm cumulative} = 0.934$.

From Selected Events to Sensitivity

- Signal and background yields are normalised to the same exposure, 1.75×10^{19} POT. Signal events are weighted from the HNLux event metadata using

$$w_{\mathrm{ref}} = P_w \, N_w \, \frac{\Gamma(P \to N)}{\Gamma_P} \, B_N \, \frac{N_{\mathrm{POT}}}{\mathrm{PoT}_f}, \qquad |U_{\alpha N}|^2_{\mathrm{ref}} = 10^{-6}.$$

P_w	parent meson/event production weight,
N_w	HNL decay-in-detector probability weight from HNLux,
$\Gamma(P \to N)$	partial width for producing the HNL from parent P,
Γ_P	total width of the parent particle,
B_N	branching fraction for the selected HNL decay mode,
N_{POT}	target exposure used for the sensitivity calculation,
PoT_f	POT normalization stored for the generated HNLux sample.

$$P_{\mathrm{dec}}(x) = e^{-L/L_{\mathrm{lab}}(x)} \left[1 - e^{-\Delta \ell_{\mathrm{det}}/L_{\mathrm{lab}}(x)}\right].$$

For each HNL mass and trial mixing value, the decay probability is recomputed event-by-event, giving a new signal weight and total selected signal yield; the 90% C.L. sensitivity is the mixing where this yield reaches the required signal threshold.

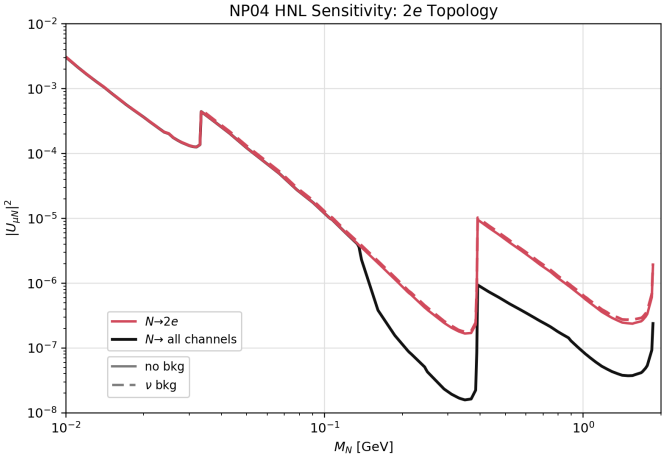
Selected yields are converted into a 90% C.L. reach

$$S_{\mathrm{req}}(B) = \max\left[2.44,\ \sqrt{4.61\,(B + (0.2B)^2)}\right].$$

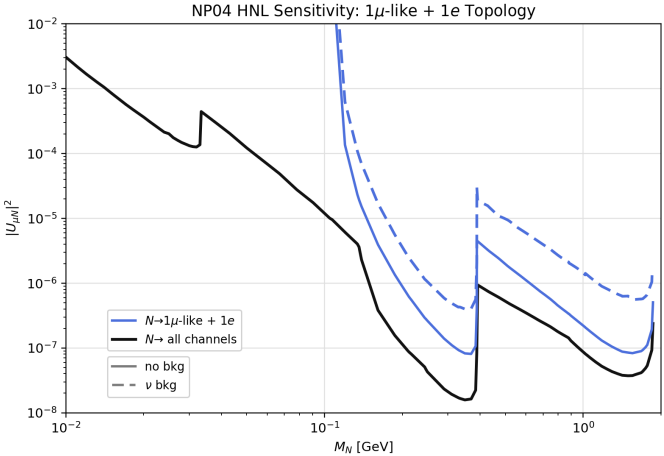
1. Recompute the full-geometry signal yield for each trial mixing.
2. Interpolate each parent-family yield in mass, then sum.
3. Solve $S(U^2) = S_{\mathrm{req}}(B)$.

Simplified one-bin Gaussian treatment following the approach of *Heavy Neutral Lepton searches at an ICARUS-like detector using NuMI beam*, arXiv:2408.03383v2; 20% background normalisation uncertainty.

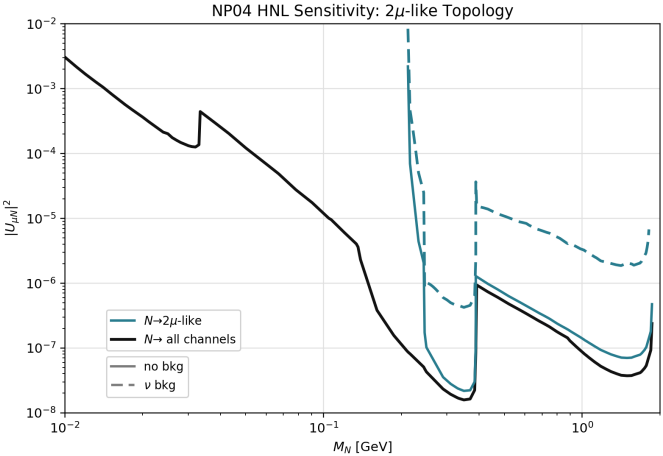
Final sensitivity: $2e$ topology



Final sensitivity: 1μ-like $+1e$ topology



Final sensitivity: 2μ-like topology



Next Steps

- Validate the full analysis chain for $|U_{eN}|^2$ mixing.
- Optimise the kinematic cuts directly for the expected 90% C.L. sensitivity, rather than for background rejection alone.
- Quantify the sensitivity gain relative to the current ϵ_S/ϵ_B-optimised selection.
- Since different mixing values might make certain energy particles more likely, I would also like to quantify mixing has a significant effect on energy cuts over the mass range.

Thank you for listening!